INCH-POUND
A-A-59806A
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COMMERCIAL ITEM DESCRIPTION

LIQUID LEVEL SWITCH, SIDE-MOUNTED, FLOAT ARM MAGNET-ACTUATED TYPE, SHIPBOARD SEWAGE AND GRAYWATER SYSTEMS

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for side-mounted, float arm magnet-actuated, liquid level switches used to indicate discrete points of fluid level.

2. CLASSIFICATION. The liquid level switch shall conform to the following types:

a. Type I – Liquid Level Switch, Regular Flange
b. Type II – Liquid Level Switch, Special Flange
c. Type III – Liquid Level Switch, Square Flange

3. SALIENT CHARACTERISTICS.

3.1 Materials. The material for liquid level switches shall be as specified herein.

3.1.1 Corrosion. Liquid level switch parts in contact with sewage or graywater shall be made of materials resistant to corrosion by these wastes. Shipboard sewage contains freshwater or seawater mixed with various concentrations of human wastes and fixture (water closet and urinal) cleaning agents, in accordance with S6480-A4-CAT-010, and with most of the cleaners belonging to the following chemical component/cleaning families:

a. Quaternary ammonium compounds (Chemical Abstracts Service [CAS] Numbers 32426-11-2, 5538-94-3, and 68424-85-1)
b. Ethoxylated alcohols
c. Dipropylene monomethyl (DPM) glycol ether (CAS Number 034590-94-8)
d. Betaines
e. Alpha olefin sulfonates
f. Alkanolamides
g. Alkanolamines
h. Hydrotropes
i. Hydroxides (potassium, sodium, and ammonium)
j. N-Alkyl dimethylamine oxides
k. Acids (sulfamic, muriatic, phosphoric, citric, glycolic, and acetic)
l. D-Limonene (CAS Number 5989-27-5)

Beneficial comments, recommendations, additions, deletions, clarifications and any other data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 6680

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m. Sodium metasilicate pentahydride (CAS Number 6834-92-0)

n. Sodium tripolyphosphate (CAS Number 7758-29-4)

o. Polyglycols

p. Urea (CAS Number 57-13-6)

q. Propylene glycol (CAS Number 57-55-6)

r. Iminoacetic acid (including ethylenediamine tetraacetic acid [EDTA])/ chelating agents

Shipboard sewage, in addition to the wastes listed above, may contain scale prevention products in accordance with MIL-PRF-32217. Graywater contains discarded water from deck drains, lavatories, showers, dishwashers, laundries, and garbage grinders, as well as discarded water from shipboard medical facilities (this does not include industrial wastes, infectious wastes, and human body wastes).

3.1.2 Alternate materials. Alternate materials may be used in lieu of those materials specified herein, provided these materials have greater corrosion resistance and improved physical and electrical qualities for the intended use than the materials specified.

3.2 Design and construction.

3.2.1 Design. Float arm magnet-actuated type liquid level switches shall employ opposing, permanent magnets to activate electric switches located in a switch module. As the float module travels to its upper and lower limits, its magnet shall repulse a magnet located in the switch module (see figure 1). As the switch module's magnet is repulsed in either direction, it shall move a mechanical linkage in the switch module that in turn changes the settings of two independent electric switches.

3.2.2 Construction. Each liquid level switch assembly shall consist of a float module, float permanent magnet, flange module, and switch module.

3.2.2.1 Float module. The float module shall consist of a cylindrical float with the dimensions shown on figure 2, shall be constructed of stainless steel alloy 316 Ti, and shall be capable of operating in fluids with 0.85 minimum specific gravity. A hole of sufficient size to accept fasteners for attachment to the float permanent magnet shall be provided.

3.2.2.2 Float permanent magnet. The float permanent magnet (see figure 3) shall be provided with a stainless steel alloy 316 Ti shaft on one end with two holes of sufficient size to accept fasteners for attachment to the float module and flange module.

3.2.2.3 Flange module. The flange module shall be constructed of stainless steel alloy 316 Ti. The flange module shall be available in three mounting flange types with dimensions and drill patterns conforming to figures 4, 5, and 6.

3.2.2.3.1 Type I (regular flange). The flange shall include six evenly spaced drilled holes with the vertical plane of the float module-operating arc straddling the drilled holes (see figure 4).

3.2.2.3.2 Type II (special flange). The flange shall include six evenly spaced drilled holes with the vertical plane of the float module-operating arc passing through the center of the drilled holes (see figure 5).

3.2.2.3.3 Type III (square flange). The flange shall include four evenly spaced drilled holes with the vertical plane of the float module-operating arc straddling the drilled holes (see figure 6).

3.2.2.4 Flange facing. The surface of the flange module in contact with the tank wall shall be a machined surface in accordance with ASME B46.1. It shall have a finish with a circular lay of 500 to 1,000 roughness average (RA) (concentric) or 125 to 500 RA (spiral), which is produced by machining 30 to 80 serrations of uniform depth per inch of face width.

3.2.2.5 Flange linkage. Dual brackets (see figure 7) constructed of stainless steel alloy 317L shall be provided on the wet side of the flange module for attachment of the float permanent magnet.

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3.2.2.6 Flange linkage cover. A cover constructed of stainless steel alloy 317L shall be provided on the wet side of the flange module. The cover shall extend over the float permanent magnet and shall not interfere with operation of the switch.

3.2.2.7 Flange boot. The open end of the flange linkage cover and float module shaft shall be fitted with a boot made of flexible, synthetic rubber (Buna-N, shore hardness 40-50) to prevent liquids and solids from entering the linkage cavity. The boot shall not interfere with operation of the switch.

3.2.2.8 Miscellaneous hardware. Fasteners (e.g., pins and clips) used on the float side of the assembly shall be constructed of stainless steel alloy 317L.

3.2.2.9 Switch module. The switch module housing shall consist of the switch module enclosure and linkage/magnet cover (see figure 8). Both the switch module enclosure and linkage/magnet cover shall be constructed of seawater-resistant, die cast aluminum in accordance with BS EN 1706. Threaded fasteners shall be provided for securing the switch module enclosure and linkage/magnet cover to the flange module. The linkage/magnet cover shall be provided with gaskets where it mates with the switch module enclosure and flange module. The wiring connection at the bottom of the switch module enclosure shall be female, ¾-inch, class 2A National Pipe Thread.

3.2.2.10 Switch module enclosure. The switch module enclosure shall contain a magnet, linkage, two independent electric switches, and a terminal board (see figure 9). A hinged cover constructed of seawater-resistant, die cast aluminum in accordance with BS EN 1706 shall be provided on the end of the switch module that provides access to the terminal board. The enclosure shall be provided with a gasket where it mates with the linkage/magnet cover and threaded fasteners for securing the cover to the enclosure.

3.2.2.11 Marking. The outside of the switch module enclosure cover shall be permanently marked or labeled with the manufacturer's part number, company name, and address. The inside of the cover shall have a connection diagram (see figure 10) etched with "NO" (for Normally Open), "NC" (for Normally Closed), or "C" (for Common) designations for switch contacts or numbers assigned by the manufacturer.

3.2.2.12 Magnet and linkage. The switch module magnet shall be affixed to the switch module linkage using threaded fasteners and installed facing the flange module.

3.2.2.13 Electric switches. The electric switches shall be single pole, double throw type micro switches with screw-type terminals, rated for 5 amps at 250 volts alternating current (VAC) and with silver contacts in accordance with NEMA 4 or IP65 standards. Each switch's terminal contacts shall be marked "NO" (for Normally Open), "NC" (for Normally Closed), or "C" (for Common) (see figure 11); or graphically with symbols (see figure 12).

3.2.2.14 Wiring and terminal board. Wiring from the electric switches to the terminal board shall be constructed of solid, corrosion-resistant wire rated for a minimum of 5 amps. The terminal board shall be a surface-mounted, one-piece molded, barrier type terminal board with six screw-type terminals rated for a minimum of 5 amps. Each terminal connection shall be permanently marked "NO", "NC", or "C", as applicable, or with numbers corresponding to those assigned by the manufacturer for the connection diagram etched on the inside cover.

3.3 Performance characteristics.

3.3.1 Environmental.

3.3.1.1 Temperature. The liquid level switch shall operate in liquids with temperature variations from 40 to 160 °F with no effect on switch operation.

3.3.1.2 Pressure. The liquid level switch and all wetted parts shall withstand, and operate subjected to, continuous external liquid pressure of 40 pounds per square inch or greater without moisture intrusion causing mechanical or electrical failures or corrosion.

3.3.2 Voltage and frequency. The liquid level switch shall operate with input voltages of 115 VAC, 60 hertz single-phase power in accordance with MIL-STD-1399-300-1.

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3.3.3 Service life. The liquid level switch shall be designed for not less than 48 months of trouble-free operation when installed aboard ship in a sewage or graywater environment (see 3.1.1).

3.3.4 Underwriters Laboratories, Inc. (UL) certification. The liquid level switch assembly shall be examined and tested for conformance to UL 508 for use in water and sewage and any other tests and inspections UL deems necessary for certification.

3.3.5 Operation.

3.3.5.1 Switch operation. The float arm magnet-actuated liquid level switch shall operate by movement of opposing magnets that activate electric switches. As the switch float rotates, the linkage attached to the float shall bring a magnet in close proximity to the non-magnetic surface of the switch flange. This shall allow a magnetic signal from the float magnet to be transmitted across the flange wall. This magnetic signal shall repulse a second magnet located on the linkage in the switch module since both magnets are positioned with like poles facing each other. The force of the repulsion on the second magnet shall cause the mechanical linkage in the switch module to move, which in turn changes the settings of two independent electric switches.

3.3.5.2 Electric switch operation. Each electrical switch shall have one NO contact, one NC contact, and one C contact. NO contacts shall be open when the end of the float is 15/32 inch below the horizontal plane and closed when the end of the float is 15/32 inch above the horizontal plane (see figure 13). NC contacts shall be closed when the end of the float is 15/32 inch below the horizontal plane and open when the end of the float is 15/32 inch above the horizontal plane.

3.4 Workmanship. The liquid level switch shall be in accordance with the specifications herein, shall be without deformation or malfunction, and shall be clean and free of cracks and burrs. Surfaces of castings, forgings, molded parts, stampings, and machined and welded parts shall be clean and free from sand, dirt, sharp edges, scales, flux, and other harmful or extraneous materials. These surfaces shall also be free of defects such as porosity, under cuts, voids, and gaps. External surfaces shall be smooth and edges shall be either beveled or rounded.

3.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the conversion requirements of ASTM SI 10 and all other requirements of this specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the conversion limitations specified in ASTM SI 10 for inch-pound units, a request may be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The item offered shall have already been sold to the Government or commercial market. A modified commercial, off-the-shelf item may be used to minimize design and engineering risk. The offeror shall certify and provide supporting documentation as to the product's serviceability and the long-term (5 years) availability of all necessary maintenance and logistics support parts and materials.

5.3 Bid samples. When specified (see 7.3), one liquid level switch shall be submitted to the Government for examination and testing. The bid sample shall be non-returnable. Switch activation (see 5.4.3), shock (see 5.4.4), vibration (see 5.4.5), and electromagnetic interference (EMI) (see 5.4.6) testing shall be performed on the bid sample, as applicable. Failure of the bid sample to meet the requirements of any of the applicable tests or examinations listed herein shall be cause for rejection of the bid. After correction of deficiencies, tests shall be repeated to the extent necessary to ascertain acceptability.

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5.4 Inspection requirements.

5.4.1 Failure. Failure of any of the tests and examinations listed herein shall be cause for rejection.

5.4.2 Visual examination. The liquid level switch shall be visually examined to determine compliance with the requirements of this description and with specified dimensions.

5.4.3 Switch activation test. When specified (see 7.3), the liquid level switch shall be mounted to a test rig fabricated to simulate a shipboard bulkhead (vertical wall) with a 3½-inch diameter opening and provided with ½-inch UNC studs to match the appropriate mounting flange drill pattern (see 3.2.2.3). The leads from each switch shall be connected to an electrical circuit with the characteristics described in 3.3.2. The float shall be allowed to hang downward by gravity and shall then be raised slowly upward. The normally open switch contacts shall activate (close) and the normally closed contacts shall deactivate (open) when the end of the float is 15/32 inch above the horizontal plane shown in figure 13. The float shall then be slowly lowered back down until the switch deactivates (opens) for a normally open switch and activates (closes) for a normally closed switch with the end of the float 15/32 inch below the horizontal plane. The exact operating points during all phases of testing shall be recorded. The cycle shall be repeated three times. Failure to activate or deactivate 15/32 inch above and 15/32 inch below the horizontal plane for either switch shall constitute failure of the test. The exact rate at which the switches are cycled shall be at the manufacturer's discretion.

5.4.4 Shock test. When specified (see 7.3), the liquid level switch shall be shock tested in accordance with MIL-DTL-901 and shall meet the requirements for grade B, class I, type A hull/deck mounted equipment. During shock testing, the liquid level switch shall be mounted to simulate shipboard installations in an empty tank and shall not be restricted from normal operation and movement.

5.4.5 Vibration test. When specified (see 7.3) and while mounted in accordance with 5.4.3, the liquid level switch shall be subjected to type I vibration testing in accordance with MIL-STD-167-1 and shall not be restricted from normal operation and movement. The liquid level switch shall not suffer any cracking of a switch casing, broken or detached part, nor any impairment of its proper operation as determined by the switch activation tests of 5.4.3, which shall be repeated after the vibration test. During the test, the switch contacts shall be monitored as specified by MIL-STD-202-310. Normally open switch contacts shall not close and normally closed switch contacts shall not open while undergoing vibration testing.

5.4.6 EMI. When specified (see 7.3), the liquid level switch shall meet the following requirements of MIL-STD-461 for surface ships: CE101, RE102, CS101, RS101, and RS103 at frequency ranges of 14 kilohertz to 10 gigahertz, E-Field level 10 (volts/meter).

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AA	**59806**	**-**	**X**	**X**	**X**
Prefix for CID	CID Number	-	Type (see code below)	EMI (see code below)	High-impact (H.I.) shock grade (see code below)

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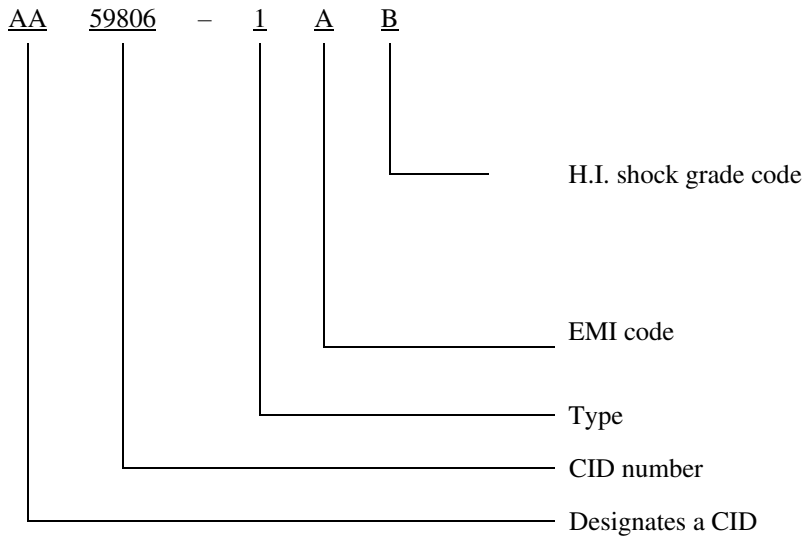
PIN Code:

Type	Code
I	1
II	2
III	3

EMI	Code
Required	A
Not required	B

H.I. shock grade	Code
Not used	A
Grade B	B
No shock required	C

Example: AA59806-1AB is for a regular flange liquid level switch (type I) that meets the EMI requirements of 5.4.6 and the grade B shock requirements of 5.4.4.



7.2 Source of documents.

7.2.1 American Society of Mechanical Engineers (ASME). ASME standards are available from the American Society of Mechanical Engineers, 150 Clove Road, Little Falls, NJ 07424-2100 or online at www.asme.org.

7.2.2 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.2.3 European standards. European standards are available from the British Standards Institution, BSI British standards, 389 Chiswick High Road, London, W4 4AL, United Kingdom, or online at https://www.bsigroup.com.

7.2.4 Defense specifications and standards. Defense specifications and standards are available online at https://quicksearch.dla.mil/.

7.2.5 Naval Sea Systems Command (NAVSEA) publications. NAVSEA publications are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. NAVSEA Publications are also available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.

7.2.6 NEMA. NEMA standards are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 900, Rosslyn, VA 22209 or online at www.nema.org.

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7.2.7 UL. UL standards are available from COMM 2000, 151 Eastern Avenue, Bensenville, IL 60106 or online at www.comm-2000.com.

7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Quantity required.

d. Type of liquid level switch (see section 2).

e. When bid samples are required (see 5.3).

f. Requirements for information to be provided with manufacturer's bid, including but not limited to sketches, catalog sheets, test records, test reports, or other documents (see 5.3).

g. Switch activation testing, if required (see 5.4.3).

h. H.I. shock testing, if required (see 5.4.4).

i. Vibration testing, if required (see 5.4.5).

j. EMI design and test requirements, if required (see 5.4.6).

k. Packaging requirements (see section 6).

l. A copy of the installation and maintenance bulletin packed with each liquid level switch assembly.

7.4 Intended use. The liquid level switches specified are intended for use in shipboard sewage Marine Sanitation Devices (MSD) and Graywater Collection and Transfer (GWCT) systems aboard surface ships or craft. An MSD is any equipment onboard a ship or craft designed to receive and treat sewage (blackwater) to a level acceptable for overboard discharge, or which receives and retains sewage onboard for later discharge ashore or in waters where discharge is permissible. The U.S. Navy uses the term MSD to also refer to equipment that receives and treats a combined sewage and graywater waste stream. The switches are used in conjunction with pump control and alarm circuitry to govern pump operation and activate tank high- and low- level alarms installed in MSDs and GWCT systems.

7.5 Bid sample inspection. Bid sample testing and examination is performed by the Government before contract award. When a bid sample is required (see 7.3), the equipment should be a production unit normally provided by the manufacturer. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and disposition of the bid samples. The manufacturer may perform bid testing; however, the Government has the right to inspect the test facilities at the manufacture plant to witness all tests specified herein. All failures, deficiencies, and discrepancies revealed during the performance of the specified tests and the corrective measures taken should be recorded and fully documented in the applicable test records and test reports.

7.6 National stock numbers (NSN). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	ITEM
6680-01-468-7292	AASPEC051
6680-01-492-1045	AASPEC12051
5930-01-497-5596	AA01051E15, 5430617
5930-01-079-9932	7000005-001
5930-01-180-3789	70372SA

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7.7 Sources of supply. Manufacturers whose products are known to meet the requirements of this CID are listed below. Competition is not limited to those products and manufacturers.

Models AASPEC051, AASPEC12051, AA01051E15
Granzow, Inc.
2300 Crownpoint Executive Drive
Charlotte, NC 28227-6702
704-845-2300

Models 5430617, 7000005-001, 70372SA
EVAC North America, Inc.
1445 Huntwood Drive
Cherry Valley, IL 61016-9560
815-654-8300

7.8 Key words.

Device, marine, sanitation

Magnetic

MSD

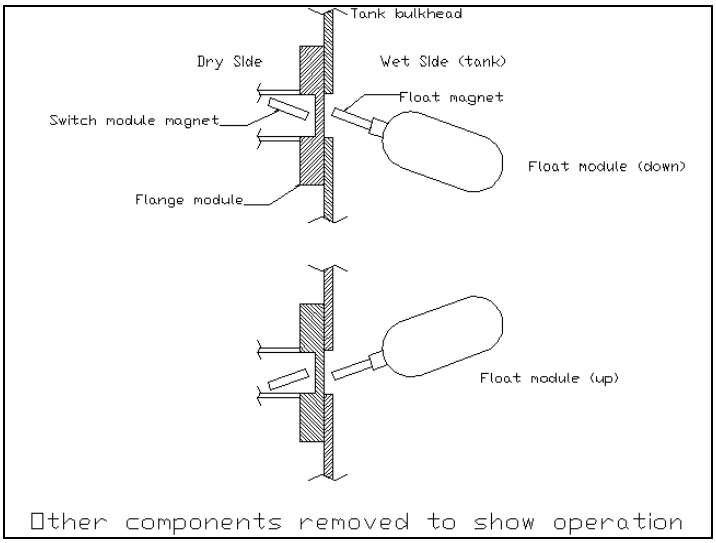


FIGURE 1. Switch module.

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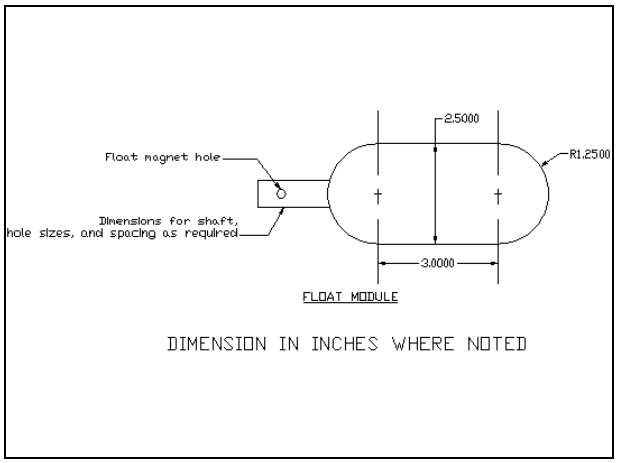


FIGURE 2. Float module.

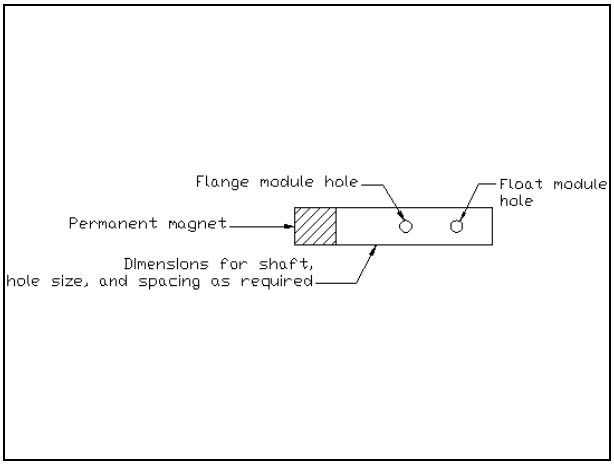


FIGURE 3. Float permanent magnet.

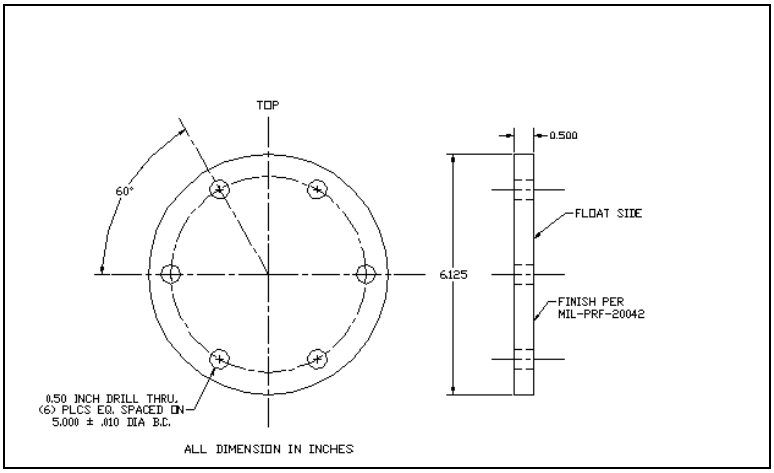


FIGURE 4. Drill pattern for type I (regular) flange module.

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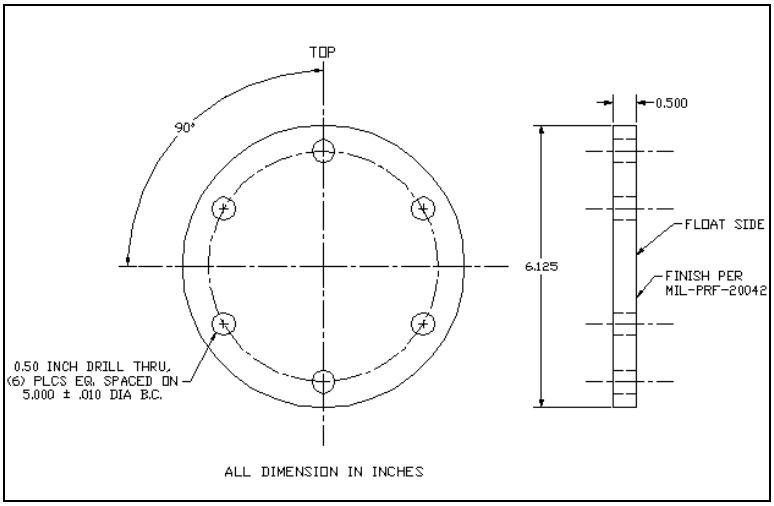


FIGURE 5. Drill pattern for type II (special) flange module.

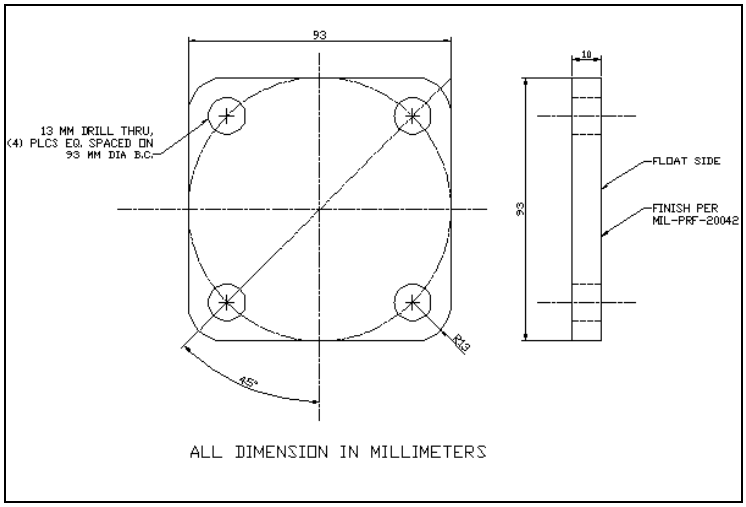


FIGURE 6. Drill pattern for type III (square) flange module.

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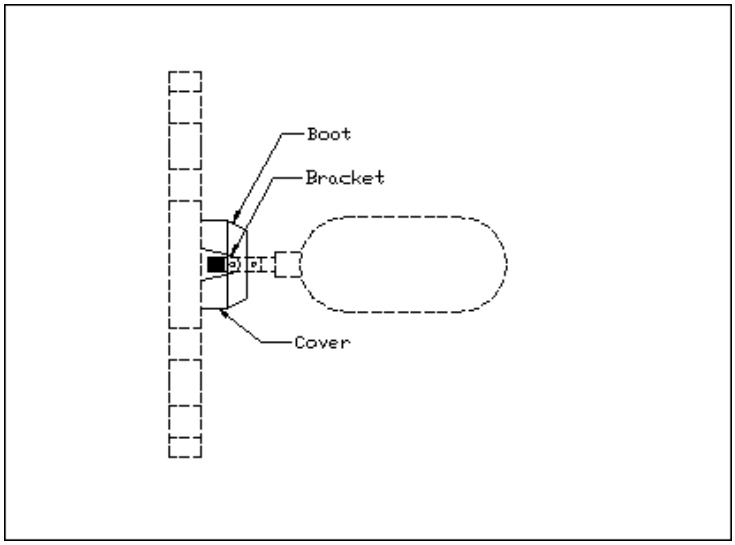


FIGURE 7. Flange linkage, cover, and boot.

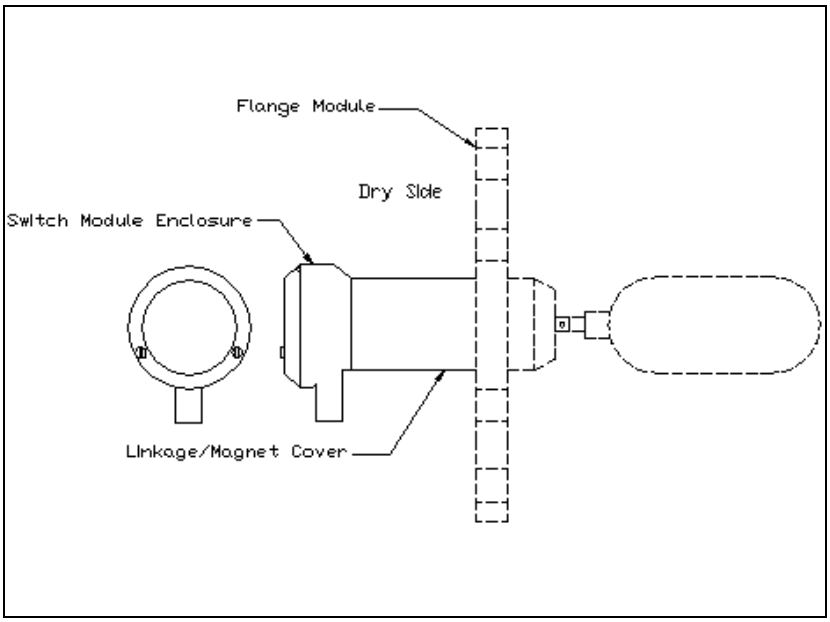


FIGURE 8. Switch module housing.

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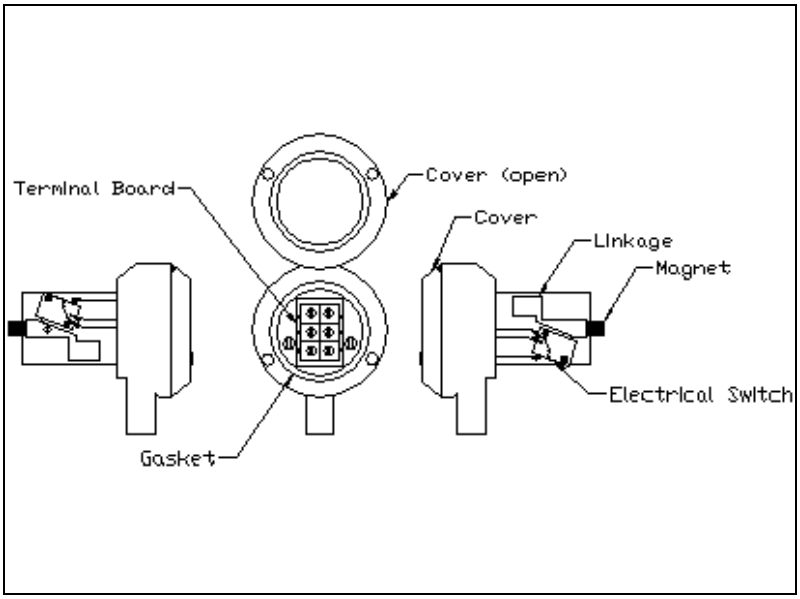


FIGURE 9. Switch module enclosure.

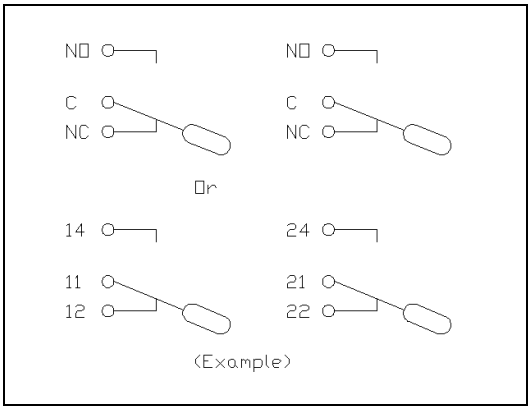


FIGURE 10. Marking.

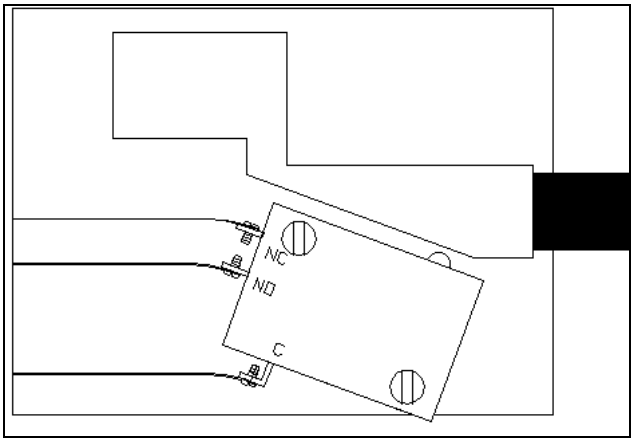


FIGURE 11. Electrical switch with NC, NO, C markings.

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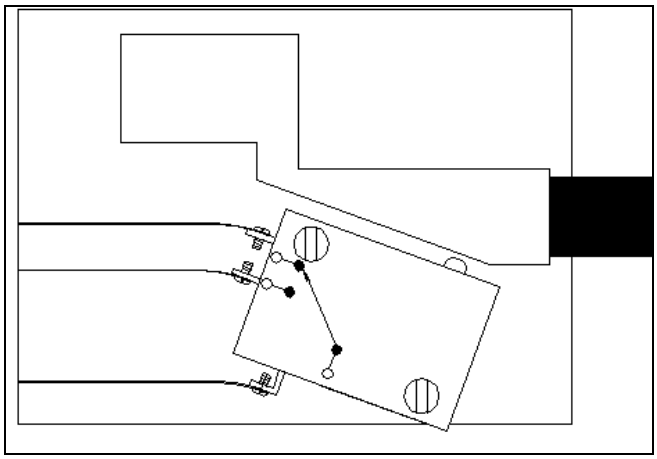


FIGURE 12. Electrical switch with graphic symbols.

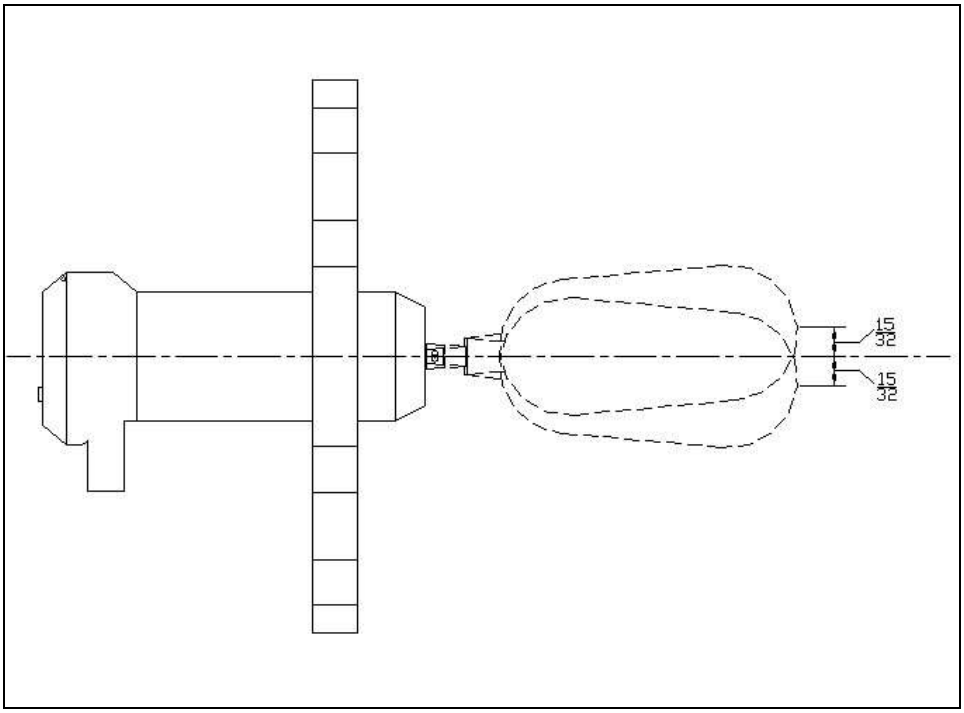


FIGURE 13. Electric switch operation.

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
- Navy – SH
- Air Force – 85
- DLA – GS

Preparing Activity:
- Navy – SH
- (Project 6680-2021-001)

Review Activities:
- Navy – AS
- Air Force – 03, 33, 71
- DLA – GS1
- CIV – NIH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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